SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS
COURSE CODE: MLA01
COURSE OUTCOME COVERED: CO1 - Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)
Assessment Tool Weightage: 45%
Total Marks: 50 **Time:** 60 Mins
No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:
I Kongara Jogesh(192525035) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: k.jogesh

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	**100%**	

Signature of the Faculty: ____________________
Total Marks Awarded: ____________

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

AI-Powered Diagnostic Imaging in Radiology: Foundation-Model Triage and Automated Reporting Systems

Domain:Healthcare

Guiding Questions Addressed in This Report

- Why this domain and application? — addressed in Sections 1, 2 and 4.
- What is the scenario? — addressed in Sections 3 and 4.
- What does the AI application contain (architecture)? — addressed in Section 7 with an architecture diagram.
- Why are these components used? — addressed in Section 7.1 (component justification table).
- Where is it used (real-world implementation)? — addressed in Section 9.
- Why is it used (impact and justification)? — addressed in Sections 5 and 10.

1. Title of AI Application

AI-Powered Diagnostic Imaging and Triage in Radiology, with specific focus on Aidoc's CARE™ foundation-model platform (body CT triage and First Read chest radiograph reporting), Viz.ai's stroke-detection suite, and NVIDIA Clara Reason for chest X-ray interpretation. The corresponding AI domain is Healthcare, specifically clinical diagnostic radiology.

This cluster of applications was chosen over a single narrow product because, together, they represent the current state of the art across the three dominant sub-categories of radiology AI in 2026: acute-condition triage (Aidoc, Viz.ai), explainable diagnostic reasoning (Clara Reason), and generative report drafting (First Read) giving a complete picture of how a modern AI-augmented radiology department actually functions rather than a single isolated feature.

2. Domain Selection

Domain selected: Healthcare , Diagnostic Radiology and Medical Imaging.

This domain was selected in preference to Finance, Retail, Autonomous Vehicles, Robotics, and Manufacturing because imaging-based AI is, by a wide margin, the most mature and evidence-rich category of applied AI in any industry today. As of March 2026, radiology alone accounted for 1,163 of the 1,524 total FDA-cleared AI medical algorithms — about 76% of all AI clearances across every medical specialty combined. This depth of regulatory, clinical-trial, and deployment data makes healthcare radiology an unusually well-documented domain for scenario-based research, in contrast to domains such as autonomous vehicles or robotics where large-scale, publicly reported real-world outcome data is comparatively sparse.

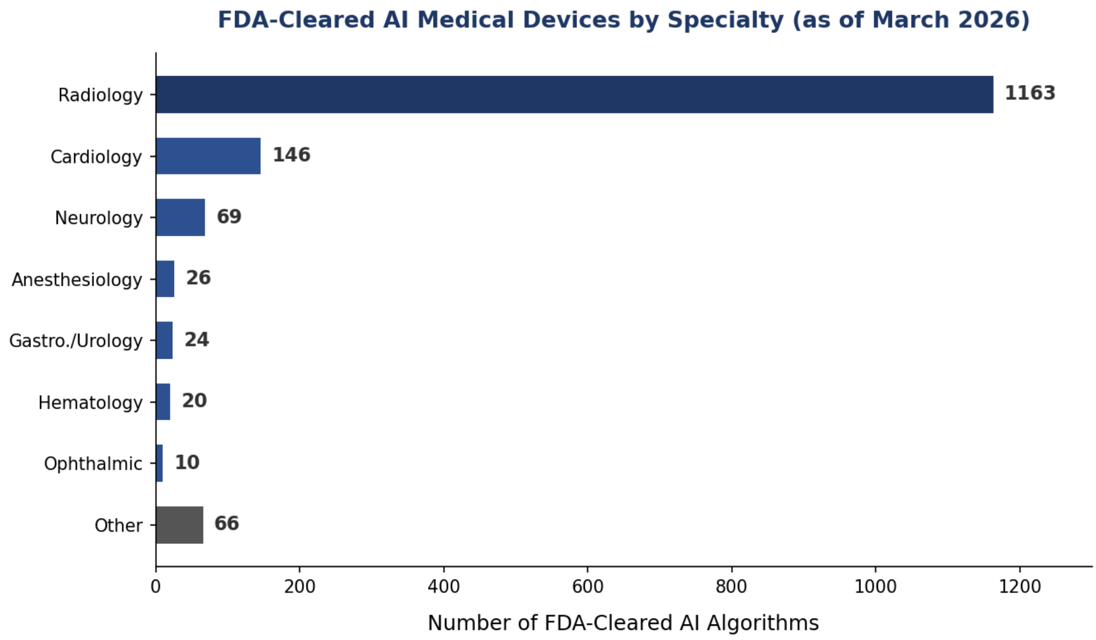


Figure 1: Distribution of FDA-cleared AI medical devices by specialty, March 2026. Radiology dominates with 76% of all clearances.

3. Introduction and Background

Radiology has become the leading clinical field for artificial intelligence adoption in medicine. As of March 2026, the FDA had cleared 1,524 AI-enabled medical algorithms, of which 1,163 belonged to radiology, with the agency now clearing roughly 30 new imaging algorithms every month, up from about 21 per month in 2024. The global market for AI in medical imaging was valued in the range of approximately USD 2–2.6 billion in 2026 by most independent analysts, with projected compound annual growth rates between 27% and 37% through the early 2030s, driven by rising imaging volumes, a global shortage of radiologists, and rapid advances in deep learning and foundation-model architectures.

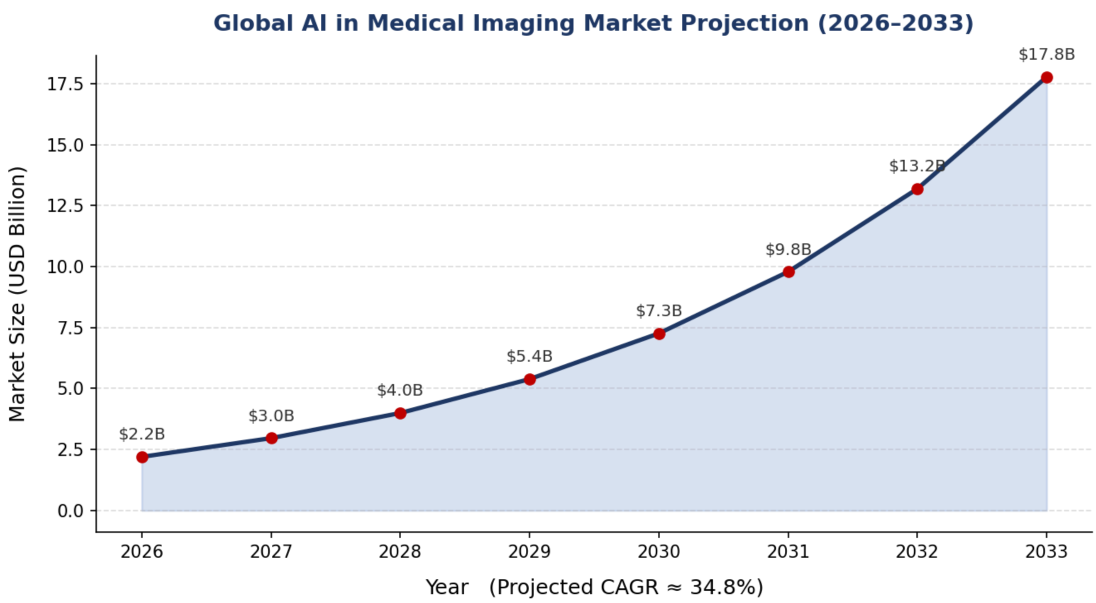


Figure 2: Projected growth of the global AI-in-medical-imaging market, 2026–2033 (illustrative CAGR-based projection derived from Coherent Market Insights base figures).

This report focuses on a cluster of closely related, currently deployed AI applications. Aidoc's CARE™ platform, in January 2026, became the first foundation-model-powered clinical AI device cleared by the FDA for whole-body CT triage across fourteen acute conditions. Its companion tool, First Read, received FDA Breakthrough Device Designation in June 2026 for drafting preliminary chest radiograph reports. Viz.ai's stroke-detection algorithms are deployed at more than 1,700 hospitals. NVIDIA's Clara Reason is an openly available vision-language foundation model that generates step-by-step diagnostic reasoning for chest X-rays rather than a bare confidence score, directly addressing the explainability problem that has historically limited clinician trust in radiology AI.

Historically, computer-aided detection (CAD) tools in radiology date back more than two decades, but were narrow, rule-based, and largely confined to mammography. The current generation differs fundamentally: it is powered by deep neural networks trained on millions of labelled images, deployed as cloud-native services integrated directly into hospital workflow software, and increasingly built on general-purpose foundation models capable of handling dozens of findings from a single scan rather than one narrow task per algorithm.

4. Problem Identification

4.1 The Scenario

The scenario this report examines is a modern tertiary-care hospital's emergency and inpatient radiology department, where imaging volumes have grown faster than the radiologist workforce. A patient arrives with acute abdominal pain or neurological symptoms; a CT or MRI scan is ordered; the resulting images must be interpreted quickly and accurately, often outside normal working hours, to guide time-critical treatment decisions such as thrombectomy for stroke or emergency surgery for bowel obstruction.

The real-world problem addressed by these applications is a widening gap between imaging demand and radiologist capacity. A Neiman Health Policy Institute study found that outpatient imaging interpretation turnaround times more than doubled between 2014 and 2023, with the steepest increases occurring in the most recent years. In emergency settings, delayed CT interpretation can add as much as 150 minutes to a patient's length of stay, directly affecting throughput and, for time-critical conditions such as stroke or aortic dissection, patient survival. At the same time, chronic disease burden and an ageing global population continue to push imaging volumes upward, while the radiologist workforce is not growing at a matching pace, resulting in fatigue, burnout, and an elevated risk of missed or delayed findings.

4.2 Why This Application Was Chosen

- It is one of the most recent (January–June 2026) FDA regulatory milestones in any AI-healthcare category, making it genuinely “latest.”

- It has verifiable, published clinical-trial performance metrics rather than only marketing claims.
- It is deployed at meaningful scale (nearly 2,000 hospitals for Aidoc; 1,700+ for Viz.ai), giving real-world, not just laboratory, evidence.
- It illustrates every major AI technique taught in this course — search/optimisation-style triage prioritisation, heuristic risk scoring, deep learning, and natural language generation — within a single coherent clinical workflow.

5. Need for AI Solution

Traditional radiology workflows process studies largely in the order they arrive, meaning a life-threatening finding such as a large-vessel-occlusion stroke or intracranial haemorrhage can sit in a queue behind routine studies. Manual interpretation is also inherently variable between readers and can be slowed by high caseloads. AI systems address these gaps in three specific ways that traditional approaches cannot easily replicate at scale.

- Automated triage: worklists are re-ordered algorithmically so the most time-critical cases are flagged and pushed to the top within minutes of acquisition — something a purely first-in-first-out human queue cannot do without extra staffing.
- Consistent pattern recognition: tireless, high-volume image analysis without the sensitivity loss that comes from human fatigue over a long shift.
- Draft report generation: shortens the time a radiologist spends on documentation, freeing more attention for complex clinical judgement rather than the mechanics of dictation.

In effect, AI functions as a heuristic pre-filter layered on top of the radiologist's own expertise — conceptually similar to how a best-first search algorithm reorders a problem space by an estimated cost function so that the most promising (here, most clinically urgent) states are explored first, rather than the plain queue-based approach used in unassisted workflows.

6. AI Technologies Used

6.1 Core Techniques

- Deep Learning / Convolutional Neural Networks (CNNs): used for classification and detection tasks such as identifying haemorrhage, fractures, or nodules on 2D and 3D imaging. CNNs learn hierarchical visual features (edges, textures, shapes, then organ-level patterns) directly from labelled scans.
- Computer Vision and Image Segmentation: used to delineate organs, lesions, and vascular structures pixel-by-pixel (e.g., NVIDIA Clara Segment for interactive 3D segmentation), which is a prerequisite for accurate volumetric measurement of tumours or bleeds.
- Vision-Language Foundation Models: large multi-modal models such as NVIDIA's NV-Reason-CXR-3B (a 3-billion-parameter model) and the architecture underlying Aidoc's CARE™ platform. These are trained on millions of image-report pairs and can generalise

across many findings from a single backbone rather than needing a separate narrow model per condition — a major shift from the one-model-per-disease paradigm of the 2018–2022 era.

- Natural Language Processing (NLP) and Generative AI: used by tools such as Aidoc's First Read and Radiology Partners' Cognita to convert visual findings into structured, human-readable draft radiology reports using large language model decoders conditioned on the image encoder's output.
- Generative Synthetic Data Models: tools such as Clara Generate produce synthetic CT and MR images to augment limited or imbalanced training datasets, particularly for rare conditions where real annotated examples are scarce.
- Cloud-based and Edge Inference Pipelines: the majority of deployments run as cloud-hosted or on-premise inference services integrated directly into hospital PACS (Picture Archiving and Communication System) and worklist software, allowing sub-minute turnaround from image acquisition to alert.

7. Working Mechanism

The pipeline begins the moment a scan (CT, MRI, or X-ray) is acquired and stored in the hospital's PACS in DICOM format. An integration layer such as Aidoc's aiOS™ automatically forwards a copy of the study to the AI inference engine in parallel with the normal workflow, so radiologists are not required to change how they order or view studies.

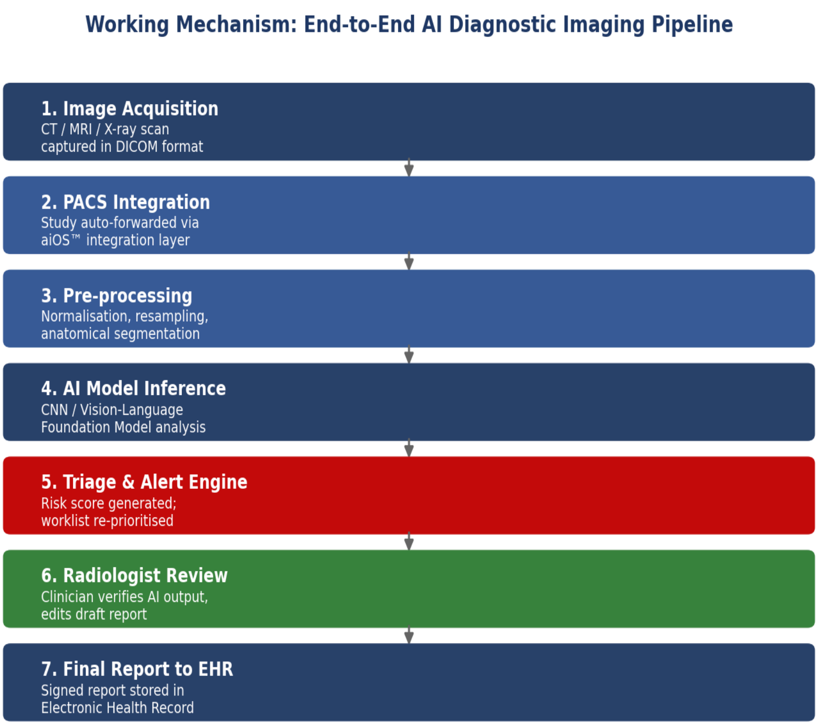


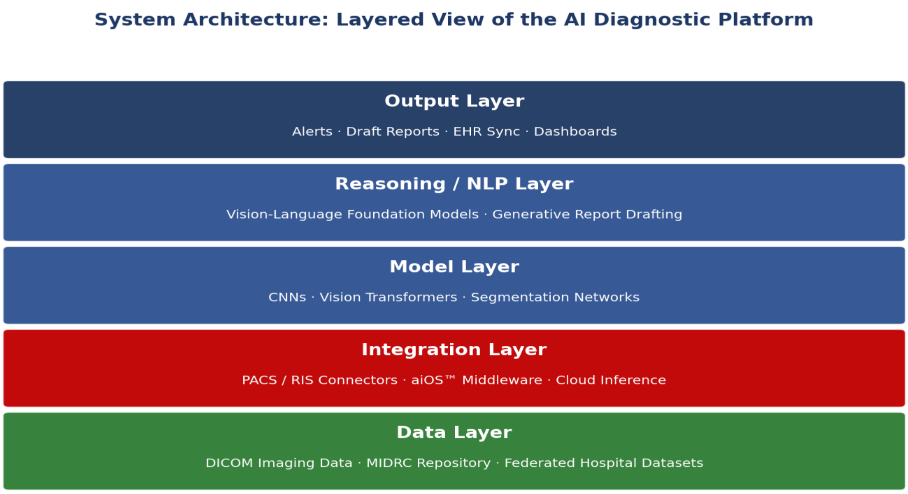
Figure 3: End-to-end working mechanism of the AI diagnostic imaging pipeline, from image acquisition to signed report.

Inside the model, the image is pre-processed (normalised, resampled, and in some cases segmented into anatomical regions) before being passed through the trained neural network. For triage tools, the network outputs a probability score for each target condition. If the probability crosses a validated clinical threshold, the system generates a real-time alert — for stroke cases this is typically pushed as a smartphone notification directly to the on-call stroke team — and re-prioritises the case at the top of the radiologist's worklist.

For reasoning- and report-oriented tools such as Clara Reason and First Read, the vision-language model additionally produces a chain of interpretable, step-by-step observations before generating draft report text. This output is always positioned as an assistive draft: the radiologist reviews, edits, and formally signs off on the final report, preserving human accountability for the diagnosis.

7.1 System Architecture and Component Justification

The platform can be understood as five layered components, shown in Figure 4. Each layer exists to solve a distinct engineering problem in the pipeline, as justified in Table 1.



Architecture Layer	Components	Why This Component Is Used
Data Layer	DICOM images, MIDRC repository, federated datasets	Provides the diverse, de-identified training and inference data required for the model to generalise across patient populations and scanner types.
Integration Layer	PACS/RIS connectors, aiOS™ middleware, cloud inference	Ensures the AI runs silently alongside existing hospital IT with zero workflow disruption for radiologists.
Model Layer	CNNs, vision transformers, segmentation networks	Performs the core visual pattern-recognition task of detecting and localising abnormal findings in the image.
Reasoning/NLP Layer	Vision-language foundation models, generative drafting	Converts raw visual detections into human-interpretable, step-by-step clinical reasoning and draft report text,

Architecture Layer	Components	Why This Component Is Used
		addressing the explainability gap of earlier CNN-only systems.
Output Layer	Alerts, draft reports, EHR sync, dashboards	Delivers actionable output directly into the clinician's existing tools (smartphone alert, worklist, EHR) so the AI insight reaches the decision-maker within minutes.

8. Data Sources and Processing

Model development relies on large, de-identified imaging repositories. In the United States, the NIH-supported Medical Imaging Data Resource Center (MIDRC) has compiled more than two million de-identified CT, X-ray, and MRI studies drawn from diverse patient populations specifically to support AI training and bias evaluation. International collaborations such as the European Imaging COVID-19 AI Database linked imaging data across 28 hospitals in 12 countries for model development.

Data processing pipelines include de-identification to strip protected health information, standardisation of image resolution and orientation, expert radiologist annotation of ground-truth labels, and, increasingly, federated learning approaches that let models train across multiple hospital datasets without the raw images ever leaving the originating institution. Because real clinical datasets are often imbalanced (rare conditions are, by definition, rare), synthetic data generation tools are used to augment training sets for underrepresented findings, improving robustness without compromising patient privacy.

9. Real-World Implementation

Aidoc's platform is deployed across nearly 2,000 hospitals worldwide, with its CARE™ foundation model receiving FDA clearance in January 2026 as the first foundation-model-powered clinical AI device, covering fourteen acute conditions including aortic dissection, appendicitis, bowel obstruction, and splenic injury from a single body CT scan. In April 2026 the company raised a $150 million Series E round led by Goldman Sachs, with NVIDIA's venture arm among the investors, taking total funding past $500 million — a strong market signal of confidence in the technology.

Viz.ai's stroke-detection algorithms, which identify large vessel occlusions from CT angiography and alert stroke teams in real time, are active at more than 1,700 hospitals across over 50 FDA-cleared algorithms. NVIDIA's Clara Reason model is being integrated by the National Institutes of Health into radiology research workflows. Elsewhere in the sector, Harrison.ai received clearance in March 2026 for acute ischaemic stroke triage software, and RadNet acquired the European AI company Gleamer for up to €230 million to expand clinical AI offerings, illustrating how implementation is scaling through both organic deployment and industry consolidation.

9.1 Comparative Snapshot of Leading Implementations

Platform	Primary Use Case	Regulatory Status (2026)	Reported Performance
Aidoc CARE™	Whole-body CT triage, 14 conditions	FDA cleared, Jan 2026	97% mean sensitivity, 98% mean specificity
Aidoc First Read	Chest X-ray report drafting	FDA Breakthrough Device, Jun 2026	Preliminary report drafting; under clinical evaluation
Viz.ai	Stroke (LVO) detection & alerting	50+ FDA-cleared algorithms	Time-to-treatment reduced by up to 52 minutes
NVIDIA Clara Reason	Chest X-ray reasoning (open model)	Research / NIH integration	Step-by-step explainable output
Paige Prostate	Pathology specimen analysis	FDA cleared	0.99 sensitivity; 65.5% faster diagnosis

10. Impact and Benefits

- Diagnostic accuracy: Aidoc's body CT triage tool demonstrated 97% mean sensitivity and 98% mean specificity across fourteen conditions in its pivotal FDA study.
- Speed to treatment: Viz.ai's stroke pathway has been shown in multicentre trials to reduce time-to-treatment for large vessel occlusion by as much as 52 minutes, a critical margin in stroke care where every minute of delay costs brain tissue.
- Workflow efficiency: Paige Prostate, a pathology-adjacent AI tool, achieved 0.99 sensitivity at the specimen level while cutting diagnosis time by 65.5%.
- Reduced turnaround pressure: automated triage and draft-reporting tools directly target the imaging turnaround bottleneck, helping offset the workforce shortage without requiring proportional increases in radiologist headcount.
- Reimbursement momentum: cardiac imaging AI became, in 2026, the first radiology AI category to receive dedicated CPT Category I reimbursement codes, giving hospitals a clearer financial pathway to fund AI adoption in at least one clinical area.
- Predictive potential: UK Biobank research using AI analysis of cardiac MRIs detected subclinical heart disease in asymptomatic individuals with roughly 88% accuracy, pointing toward a shift from reactive diagnosis to preventive screening.

11. Challenges and Limitations

Despite strong performance metrics, large-scale deployment of AI in radiology has surfaced real and unresolved risks, summarised visually in the SWOT analysis below and discussed in detail thereafter.

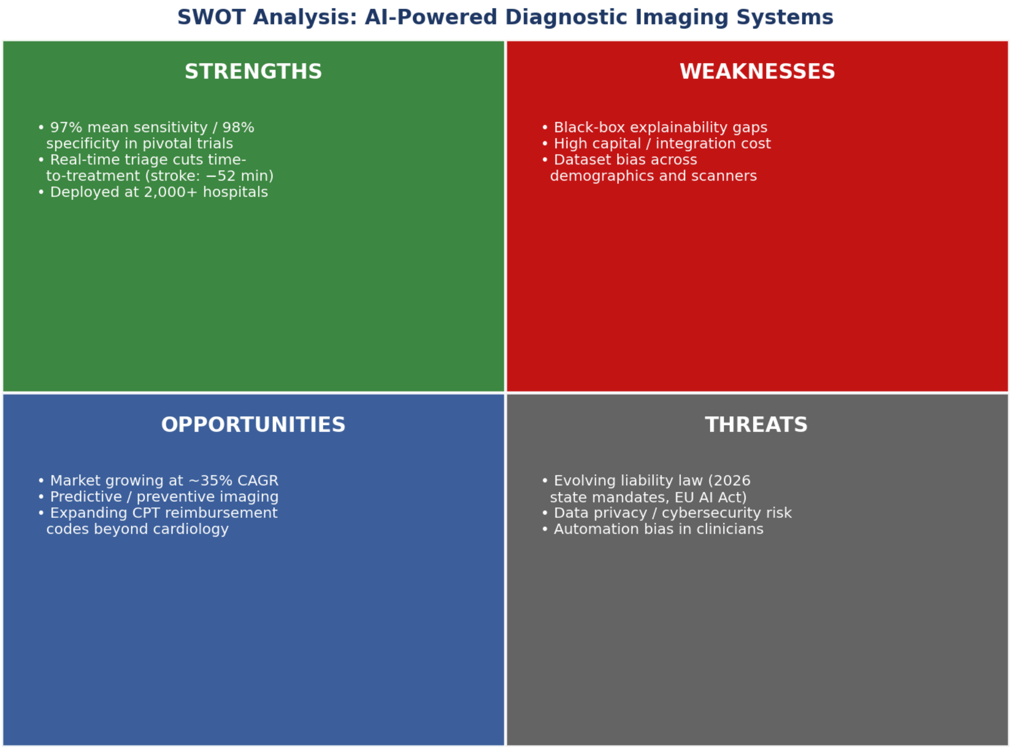


Figure 5: SWOT analysis of AI-powered diagnostic imaging systems.

11.1 Detailed Discussion

- Algorithmic bias: multiple 2025–2026 reviews document bias risks in breast- and chest-imaging AI datasets, since models trained predominantly on data from certain demographics or scanner types may underperform on underrepresented populations.
- Explainability (the "black-box" problem): many high-performing deep learning models still cannot fully justify individual predictions, which limits clinician trust and complicates error review — a gap that reasoning-oriented models like Clara Reason are specifically designed to narrow.
- Liability and legal ambiguity: the ECRI patient safety organisation named the AI diagnostic dilemma its top patient-safety concern for 2026, warning that AI is not infallible and can itself contribute to diagnostic error, while several U.S. states (including California, Texas, Alabama, Indiana, Utah, and Washington) have introduced 2025–2026 laws requiring human review and disclosure of AI use in diagnosis, without shifting legal responsibility away from clinicians.
- Regulatory complexity: the EU AI Act, taking full effect through 2026, classifies medical imaging AI as "high-risk," mandating bias audits, representative datasets, and continuous post-market surveillance — adding compliance overhead alongside FDA requirements in the United States.
- Cost and access barriers: capital costs restrict adoption particularly among smaller or rural facilities, and only a minority of hospitals currently have reimbursement pathways covering AI-assisted reads outside cardiac imaging.

- Data privacy and cybersecurity: training and deploying models on sensitive patient imaging increases exposure to data breaches and raises HIPAA/GDPR compliance complexity.
- Automation bias: over-reliance on AI output by time-pressured clinicians risks eroding independent critical judgement, a concern explicitly flagged by patient-safety bodies in 2026.

12. Future Scope and Emerging Trends

12.1 Toward General-Purpose Foundation Models

The field is moving from narrow, single-finding algorithms toward general-purpose foundation models capable of covering dozens of conditions from a single scan, as demonstrated by Aidoc's fourteen-condition body CT clearance. Future clearances are likely to extend this pattern to whole-body multi-modality coverage rather than modality-specific point solutions.

12.2 Generative Reporting at Scale

Generative AI is expanding beyond image analysis into automated report drafting (First Read, Cognita), which is likely to become a major growth area given its direct impact on radiologist workload. As these tools mature, expect tighter integration with speech and structured-data capture so that a complete, clinically-reviewed report can be produced within seconds of image acquisition.

12.3 Predictive and Preventive Imaging

Predictive and preventive imaging is emerging as a frontier: UK Biobank research has shown AI analysis of cardiac MRIs can detect subclinical heart disease in asymptomatic individuals with roughly 88% accuracy, and oncology bodies suggest that integrating such predictive analytics into screening programmes could meaningfully reduce late-stage cancer diagnoses.

12.4 Governance and Regulatory Maturation

Closer integration between AI triage systems and electronic health records, wider adoption of federated learning to train models across hospitals without centralising sensitive data, expansion of reimbursement codes beyond cardiology, and continued regulatory maturation under frameworks such as the EU AI Act and evolving FDA predetermined change control pathways are all likely to shape how quickly and safely this technology scales globally, especially into under-resourced regions with limited radiologist coverage.

13. Conclusion

AI-powered diagnostic imaging has moved from experimental promise to routine clinical infrastructure, with radiology now accounting for the large majority of all FDA-cleared AI medical devices. Platforms such as Aidoc's CARE™ and First Read, Viz.ai's stroke-detection suite, and foundation models like NVIDIA's Clara Reason demonstrate measurable gains in diagnostic sensitivity, time-to-treatment, and workflow efficiency, directly addressing the real-world problem of a radiologist shortage colliding with rising imaging demand.

At the same time, the technology's rapid scaling has surfaced serious and unresolved challenges around bias, explainability, legal liability, and equitable access that regulators, health systems, and clinicians are only beginning to formalise through frameworks such as the EU AI Act and new state-levels

14. References

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